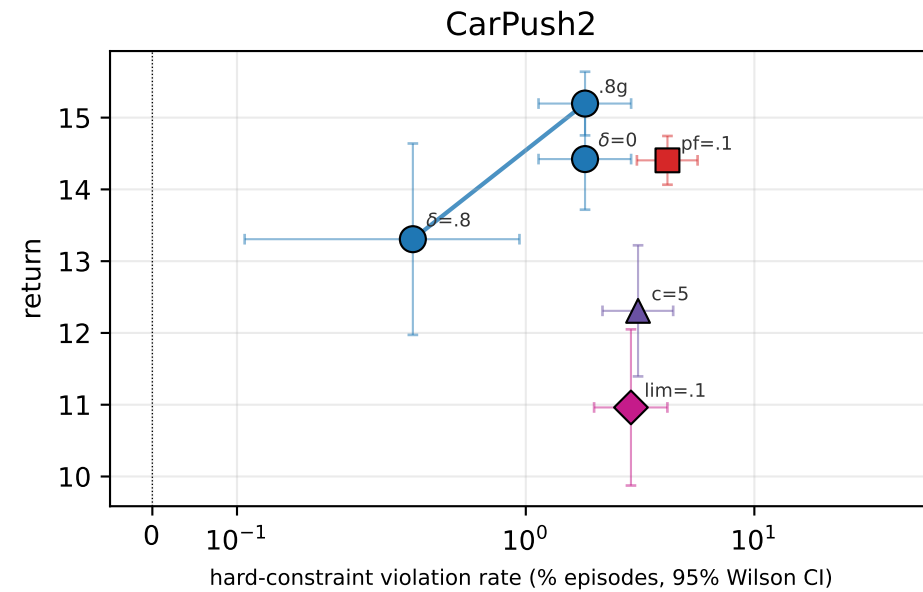
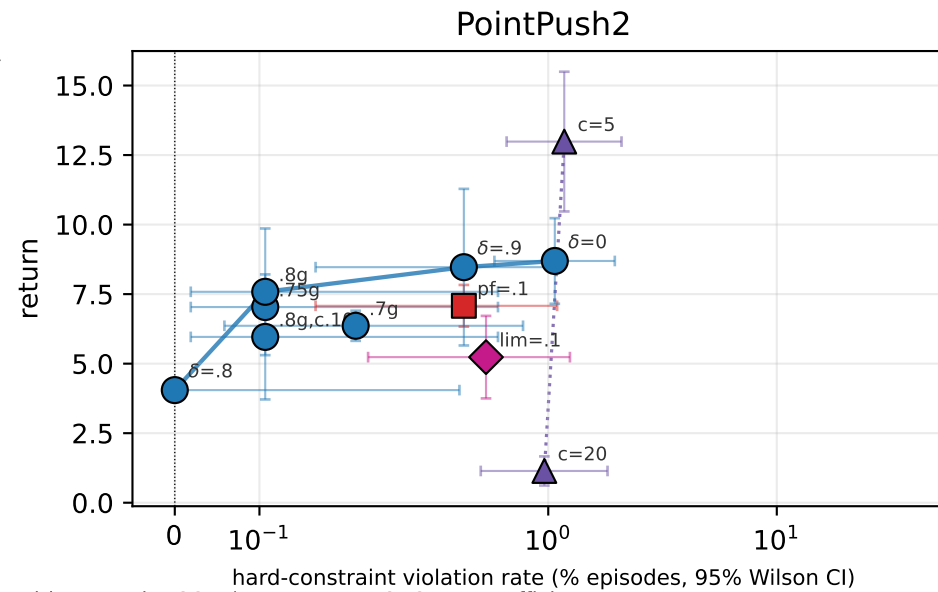
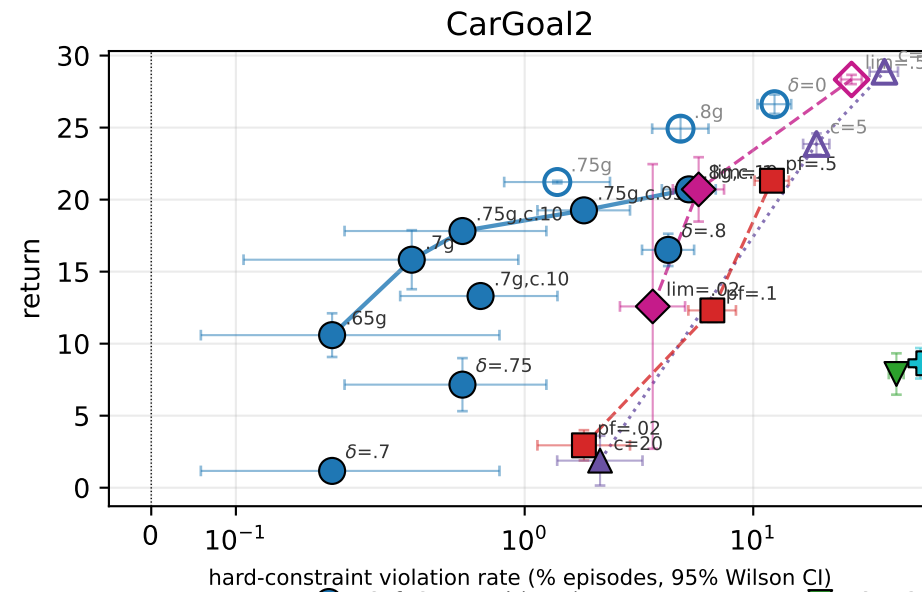
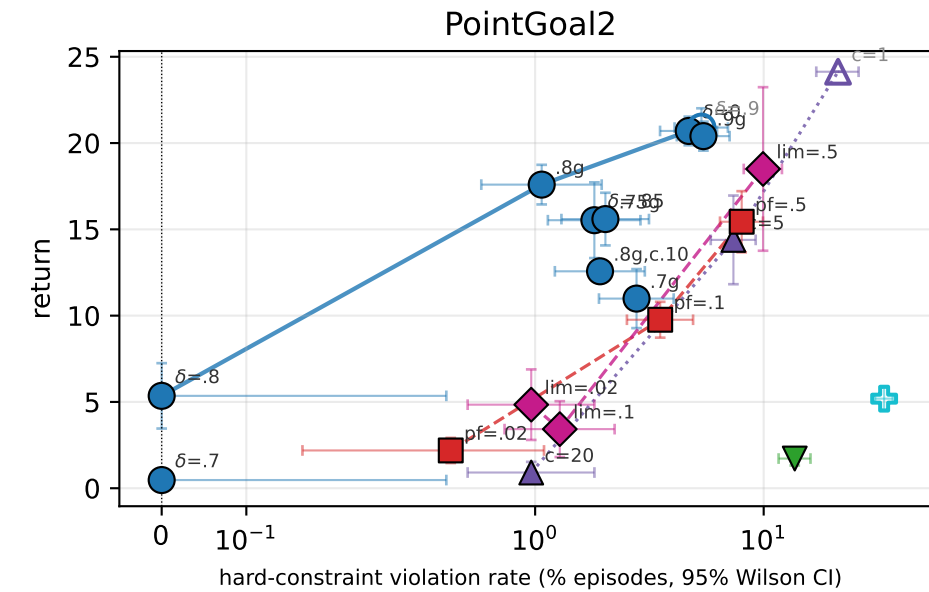


Return vs. violation rate (final checkpoint, 312 episodes/seed; hollow = soft cost > 25/episode)



- SafeOT-typed (ours)
- FDPI (constraint-based)
- ◆ SAC-Lagrangian (penalty-based)
- ▲ SAC + fixed penalty (penalty-based)

 SDAC (multi-constraint SOTA)  SAC-Pen coefficient sweep
 SRCPO (multi-constraint SOTA)  FDPI p_f sweep
 typed Pareto frontier (soft ≤ 25)  SAC-Lag cost-limit sweep